

Stereo Camera-Based Environment Perception for Automated Vehicles

Stereokamera-Basierte Umfeldwahrnehmung für automatisierte
Fahrzeuge



Scientific work for obtaining the academic degree
Master of Science (M.Sc.)
at the Department Mobility Systems Engineering
of the TUM School of Engineering and Design
at the Technical University of Munich

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Submitted on	July 01, 2025

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Submission date: October 1, 2021

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Formula Symbols

Formula Symbols	Unit	Description
B_{Fzg}	m	Breite des Fahrzeuges
B_{Rad}	m	Reifenbreite (Breite der Lauffläche)
D_{Rad}	m	Raddurchmesser
H_{Fzg}	m	Höhe des Fahrzeuges
R_{Fzg}	m	Radstand des Fahrzeuges
S_{Fzg}	m	Spur des Fahrzeuges
a_i	-	Gewichtungsfaktoren für die Kostenfunktion J
$\mathbf{c}_{\text{DGL}}$	-	Gleichungsnebenbedingungen, die aus der Diskretisierung der Zustandsdifferentialgleichung resultieren
H	-	Hamilton"-Funktion
E_{pot}	J	Potenzielle Energie
E_m	J	Spezifische Energie

Todo list

1 Introduction and Motivation

Autonomous vehicles are rapidly becoming a reality. However, their perception systems still face significant challenges in complex and unforeseen scenarios, often leading to disengagements, where human intervention via teleoperation is required. In such critical moments, the vehicle must rely on an accurate and robust perception system to ensure safe navigation.

Most existing perception systems, responsible for detecting and tracking objects in the vehicle's surroundings, rely heavily on learning-based solutions. While these methods have demonstrated impressive performance in structured and controlled environments, they often struggle to generalize in novel and untrained scenarios. This limitation is particularly problematic in safety-critical situations, where the vehicle must make quick decisions based on the perceived environment.

This project aims to develop a deterministic, stereo camera-based perception system for autonomous vehicles, capable of providing a reliable and robust understanding of the vehicle's surroundings, even in highly dynamic and unforeseen scenarios. The system will take as input two images from a stereo camera mounted on the vehicle and output a list of detected objects in the scene, along with their criticality scores, forming a situational awareness module to assess potential hazards. Additionally, this deterministic approach could run in parallel with existing learning-based perception systems, providing an additional layer of redundancy and security in critical situations where the learning-based methods may fail.

To achieve accurate 6D motion estimation (position + velocity), the system will implement a deterministic algorithm inspired by the Daimler 6D Vision system. This approach uses Kalman Filters to estimate a 6D state vector for selected feature points in the scene, which will then be clustered to identify moving objects. Unlike learning-based methods, this approach does not require extensive training data and can generalize well in new unseen scenarios.

Given the real-time requirements of autonomous vehicles, the system will be implemented in C++ using the Robot Operating System (ROS) framework. It will also incorporate GPU hardware acceleration using CUDA to speed up computationally intensive tasks, such as feature point detection, clustering or optical flow computation.

To ensure scalability and modularity, the entire system will be containerized using Docker, facilitating deployment across different hardware configurations. Finally, the developed system will be deployed and evaluated on the EDGAR research vehicle. Here, the system's performance will be compared against the state-of-the-art learning-based perception systems currently used in the vehicle.

2 State of Science and Technology

Autonomous and automated vehicles must have a comprehensive and accurate understanding of their surroundings at all times to ensure safe and reliable operation. This process, known as environment perception, is responsible for detecting, identifying, and tracking objects such as other vehicles, cyclists, pedestrians or other road elements. The quality of the measurement of the state of these objects directly influence the vehicle's ability to make safe and informed driving decisions.

Failures in the perception can lead to hazardous situations, such as collisions or unexpected navigation decisions. This is particularly critical in high-speed scenarios or complex urban environments, where there is a high density of high dynamic objects which make more complex the operation in real time.

A robust perception system must provide the following capabilities:

1. **Object detection:** Identify and locate objects in the vehicle's surroundings, such as vehicles, pedestrians, cyclists, or other road elements.
2. **Motion estimation:** Determining the position and velocity of moving objects
3. **Scene understanding:** Classifying the detected objects and assessing their criticality for the vehicle's navigation
4. **Reliability in edge cases:** Ensuring correct operation in complex and unforeseen scenarios, such as occlusions, adverse weather conditions, or sensor failures.
5. **Real-time operation:** Providing timely and accurate information to the vehicle's control system to make informed driving decisions.

While the perception concept includes different aspects, this work will focus specifically on the detection of moving objects in the vehicle's surroundings and the estimation of their 6D motion state (position and velocity) using a deterministic approach. In the following sections, we will provide an overview of the state-of-the-art of the different sensors and algorithms used in the perception systems of autonomous vehicles, focusing on the deterministic image based approaches.

2.0.1 Perception Techniques in Autonomous Vehicles

The perception systems of autonomous vehicles rely on a combination of different sensors to provide a comprehensive understanding of the ego vehicle's surroundings. In this section we will provide an overview of the most common sensor types used in autonomous vehicles and their advantages and limitations.

LiDAR-Based Perception

LiDAR (Light Detection and Ranging) sensors are increasingly used in the automotive industry for obstacle detection, 3D mapping and lane detection [1]. More and more companies are adopting this type of sensor due

to its advantage in rapidly and accurately estimating 3D distances. Examples include BMW [2], Mercedes-Benz [3], and Waymo [4].

Working principle LiDAR sensors use the concept of time of flight to measure the distance to objects in the vehicle's surroundings. The sensor emits a laser beam and measures the time it takes for the light to reflect back to the sensor. By calculating the time difference, the sensor can determine the distance to the object based on the speed of light [5].

Advantages and Limitations LiDAR sensors have the advantage of providing an accurate 3D representation of the vehicle's surroundings. This makes them particularly useful for object detection and localization tasks. However, LiDAR sensors are sensitive to adverse weather conditions, such as rain, snow or fog since the laser beams can be scattered by the particles in the air. Additionally, LiDAR sensors are more expensive than other sensor types, such as cameras or radars, which can limit their adoption in mass-market vehicles, although their prices are steadily decreasing [1].

Radar-Based Perception

Radar (Radio Detection and Ranging) use radio waves to detect objects in the vehicle's surroundings. They are commonly used in the automotive industry for object avoidance.

Working principle Similar to LiDAR sensors, radar sensors emit radio waves and measure the time it takes for the waves to reflect back to the sensor. By calculating the time difference, the sensor can determine the distance to the object based on the speed of light. Radar sensors can also estimate the object's velocity based on the Doppler effect [6].

Advantages and Limitations Radar sensors are less sensitive to adverse weather conditions than LiDAR sensors, making them more reliable in rain, snow, or fog. [1, 7]. However, radar sensors have a lower resolution than LiDAR sensors, which can limit their ability to detect small objects or provide detailed information about the object's shape.

Image-Based Perception

Image based perception systems use cameras to capture images of the vehicle's surroundings and are the principal sensor in some companies like Tesla [8]. These sensors can be used as single cameras for object detection and classification or in pairs to provide stereo vision, which allows for depth estimation and 3D reconstruction of the scene. These type of sensors can also be used for optical flow computation and the ego motion estimation through visual odometry

Monocular vision Monocular vision systems use a single camera to capture images of the vehicle's surroundings. These systems are commonly used for object detection and classification tasks, such as detecting pedestrians, cyclists, or other vehicles [1]. Monocular vision systems are less expensive than stereo vision systems and can be easily integrated into existing vehicles. However, they have limitations in depth perception and 3D reconstruction, which can make it challenging to estimate the distance to objects accurately.

Stereo vision Stereo vision systems use two cameras to capture images of the vehicle's surroundings and use the pinhole camera model [9] to estimate the depth of objects in the scene. By comparing the images from the two cameras, the system can calculate the disparity between corresponding points in the images and use this information to estimate the distance to the objects.

Advantages and Limitations Image-based perception systems have the advantage of providing a rich and detailed information about the environment. Cameras can capture color information, texture, and shape of objects, which can be useful for object detection and classification tasks. In addition, cameras are less expensive than LiDAR sensors and can be easily integrated into existing vehicles. However, image-based perception systems are sensitive to adverse weather conditions, such as rain, snow, or fog, which can affect the quality of the images and limit the system's performance. The algorithms used for object detection and tracking in image-based perception systems can also be computationally intensive, which can affect the system's real-time performance.

In the following table a summary of the advantages and limitations of the different sensor types used in autonomous vehicles is provided:

Table 2.1: Summary of sensor types used in autonomous vehicles

Sensor type	Advantages	Limitations
LiDAR	Accurate 3D representation	Expensive and sensitive to adverse weather conditions
Radar	Works in bad weather, detects velocity	Low spatial resolution
Cameras	Rich scene details, color information, Cheaper than the other considered sensors	Depth perception is limited, affected by lighting, expensive computation cost

While sensors provide raw data about the environment, perception algorithms are responsible for interpreting this information to detect and track moving objects. These algorithms can be broadly classified into learning-based and deterministic approaches, as discussed in the following sections.

2.0.2 Algorithms for Moving Object Detection

The perception algorithms can also be classified into two main categories: learning-based and deterministic approaches. Learning-based approaches use machine learning techniques, such as deep learning, and are data driven. In the other side, deterministic approaches use mathematical models to estimate the state of the objects in the scene. In the following sections we will provide an overview of the most common algorithms used for moving object detection in autonomous vehicles.

Learning-Based Approaches

Learning-based approaches have gained popularity in recent years. These methods use convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformers to process the data coming from cameras, LiDARs, and radar systems. In the work done by Wen and Jo [10], they provide a comprehensive review of the different learning-based approaches used in autonomous vehicles, which can be summarized in the following figure:

Some limitations of learning-based approaches are the need for extensive training data, which can be expensive and time-consuming to collect. These methods can also struggle to generalize in novel and untrained scenarios, which can be problematic in safety-critical situations where the vehicle must make quick decisions based on the perceived environment.

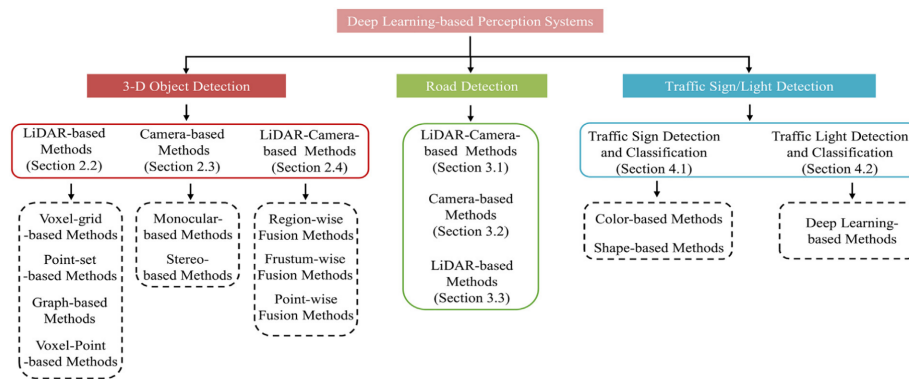


Figure 2.1: Learning-based perception methods. Source: [10]

Deterministic Approaches

Deterministic approaches use mathematical models to estimate the state of the objects in the scene.

Optical flow Optical flow is a technique used to estimate the motion of objects in the scene by tracking the movement of pixels between consecutive frames. This method is based on the assumption that the intensity of a pixel remains constant over time, which allows the system to track the movement of objects based on the changes in pixel intensity [11]. Some of the most common optical flow algorithms are the Lucas-Kanade method [12] and the Horn-Schunck method [13].

6D Vision 6D vision is a deterministic perception framework that combines stereo vision and motion tracking to estimate the 3D position and velocity of objects in the scene. Originally introduced by Franke et al [14] this method provides a direct motion estimation without requiring object segmentation. The key principles of 6D Vision are:

1. **Stereo disparity estimation:** Calculate the disparity between corresponding points in the stereo images to estimate the depth of objects in the scene.
2. **Optical flow tracking:** Track the movement of feature points in the scene between consecutive frames.
3. **Kalman filtering:** Use Kalman filters to estimate the 6D state vector (position and velocity) of the feature points in the scene.

2.0.3 State of the Art in the FTM

The Institute of Automotive Technology (FTM) at the Technical University of Munich has conducted previous research in the field of environment perception for autonomous and automated vehicles. This section provides an overview of previous work carried out at the FTM, particularly regarding the EDGAR research vehicle and the development of deterministic perception systems based on 6D vision.

2.0.4 EDGAR Research Vehicle

The EDGAR (Excellent Driving Garching) research vehicle is a state-of-the-art autonomous driving platform developed at FTM [15]. It serves as a testbench for autonomous driving research, including perception, planning, control, and sensor fusion. The vehicle is equipped with a multi-sensor setup, comprising:

1. **Stereo and monocular cameras:** For image-based perception and object detection
2. **LiDAR sensors:** For 3D mapping and obstacle detection
3. **Radar sensors:** For object avoidance and velocity estimation
4. **GPS/IMU:** For localization and motion tracking

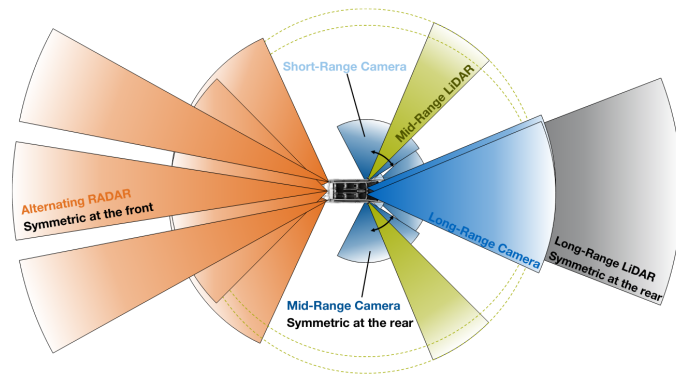
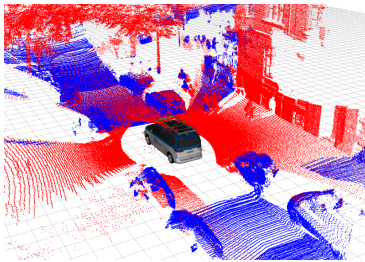


Figure 2.2: EDGAR Vehicle Sensors

2.0.5 Previous Work in the FTM

This project builds upon previous work conducted at the FTM in the field of environment perception for automated vehicles with 6D vision. Some of the key contributions include:

1. **Objektdetektion und Bewegungsprädiktion mittels eines Stereokamerasystems beim teleoperierten Fahren** [16]: Bauer in his master's thesis developed a deterministic system for object detection based on the 6D vision of Daimler [14]. The system was implemented in C++ using the ADTF framework Automotive Data and Time-Triggered Framework and was capable of detecting and tracking objects in the vehicle's surroundings with some real time limitations.
2. **Weiterentwicklung und echtzeitfähige Umsetzung eines Stereo-Vision-basierten Bewegungsprädiktionssystems für die aktive Sicherheit von teleoperierten Fahrzeugen** Some time later in [17] Rotar continued Bauers work by implementing the system in GPU using CUDA to speed up the computationally intensive tasks.
3. **Validation of an Obstacle Detection Approach for Teleoperation of Automated Vehicles:** Finally Pastor [18] started porting the system to ROS1 to facilitate the integration with other modules in the EDGAR vehicle. The system was implemented without CUDA GPU acceleration and was not deployed in the EDGAR vehicle. The system also has some problem with the object detection.

Building upon these previous contributions, this project aims to further improve deterministic perception by addressing key limitations found in prior implementations. In particular, this work will focus on enhancing computational efficiency, real-time feasibility, and system deployment within ROS2. These improvements will be detailed in the next section on objectives.

3 Methodology

4 Discussion

5 Summary

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Bibliography

- [1] J. Van Brummelen, M. O'Brien, D. Gruyer and H. Najjaran, "Autonomous vehicle perception: The technology of today and tomorrow," *Transportation Research Part C: Emerging Technologies*, vol. 89, pp. 384–406, 2018, DOI: 10.1016/j.trc.2018.02.012.
- [2] BMW. "Level 3 highly automated driving available in the new BMW 7 Series from next spring." en. 2023. Available: <https://www.press.bmwgroup.com/global/article/detail/T0438214EN/level-3-highly-automated-driving-available-in-the-new-bmw-7-series-from-next-spring?language=en> [visited on 02/26/2025].
- [3] Mercedes-Benz. "Drive Pilot," 2025. Available: <https://www.mercedes-benz.de/passengercars/technology/drive-pilot.html> [visited on 02/26/2025].
- [4] Waymo. "Meet the 6th-generation Waymo Driver: Optimized for costs, designed to handle more weather, and coming to riders faster than before," 2024. Available: <https://waymo.com/blog/2024/08/meet-the-6th-generation-waymo-driver> [visited on 02/26/2025].
- [5] Y. Li and J. Ibanez-Guzman, "Lidar for Autonomous Driving: The Principles, Challenges, and Trends for Automotive Lidar and Perception Systems," *IEEE Signal Processing Magazine*, vol. 37, no. 4, pp. 50–61, 2020, DOI: 10.1109/MSP.2020.2973615.
- [6] J. Leonard, J. How, S. Teller, M. Berger, S. Campbell, et al., "A perception-driven autonomous urban vehicle," *Journal of Field Robotics*, vol. 25, no. 10, pp. 727–774, 2008, DOI: 10.1002/rob.20262.
- [7] R. H. Rasshofer and K. Gresser, "Automotive Radar and Lidar Systems for Next Generation Driver Assistance Functions," in *Advances in Radio Science*, 2005, pp. 205–209, DOI: 10.5194/ars-3-205-2005.
- [8] K. N. Maanyu, D. G. Raj, R. V. Krishna and D. S. B. Choubey, "A Study on Tesla Autopilot," vol. 71, no. 171, 2020.
- [9] R. Szeliski, *Computer Vision: Algorithms and Applications*, (Texts in Computer Science), Cham, Springer International Publishing, 2022, ISBN: 978-3-030-34371-2 978-3-030-34372-9. DOI: 10.1007/978-3-030-34372-9.
- [10] L.-H. Wen and K.-H. Jo, "Deep learning-based perception systems for autonomous driving: A comprehensive survey," *Neurocomputing*, vol. 489, pp. 255–270, 2022, DOI: 10.1016/j.neucom.2021.08.155.
- [11] A. Alfarano, L. Maiano, L. Papa and I. Amerini, "Estimating optical flow: A comprehensive review of the state of the art," *Computer Vision and Image Understanding*, vol. 249, p. 104160, 2024, DOI: 10.1016/j.cviu.2024.104160.
- [12] S. Lucey, R. Navarathna, A. B. Ashraf and S. Sridharan, "Fourier Lucas-Kanade Algorithm," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 35, no. 6, pp. 1383–1396, 2013, DOI: 10.1109/TPAMI.2012.220.
- [13] B. K. P. Horn and B. G. Schunck, "Determining optical flow," *Artificial Intelligence*, vol. 17, no. 1, pp. 185–203, 1981, DOI: 10.1016/0004-3702(81)90024-2.

- [14] U. Franke, C. Rabe, H. Badino and S. Gehrig, "6D-Vision: Fusion of Stereo and Motion for Robust Environment Perception," in *Pattern Recognition*, 2005, pp. 216–223, ISBN: 978-3-540-31942-9. DOI: 10.1007/11550518_27.
- [15] P. Karle, T. Betz, M. Bosk, F. Fent, N. Gehrke, et al. "*EDGAR: An Autonomous Driving Research Platform – From Feature Development to Real-World Application*," arXiv:2309.15492 [cs]. Jan. 2024. DOI: 10.48550/arXiv.2309.15492.
- [16] S. Bauer, "Objektdetektion und Bewegungsprädiktion mittels eines Stereokamerasystems beim teleoperierten Fahren," MA thesis, TUM, Munich, 2014.
- [17] A. Rotar, "Weiterentwicklung und echtzeitfähige Umsetzung eines Stereo-Vision-basierten Bewegungsprädiktionssystems für die aktive Sicherheit von teleoperierten Fahrzeugen," MA thesis, TUM, Munich, 2016.
- [18] C. Pastor, "Validation of an Obstacle Detection Approach for Teleoperation of Automated Vehicles," BSc. Thesis, TUM, 2021.

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A Chapter Anhang

A.1 Section Anhang

A.1.1 Subsection Anhang

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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B Chapter Anhang 2



Figure B.1: Bild im Anhang

Table B.1: Tabelle im Anhang

Spalte 1	Spalte 2
...	...